

Candidate Name

Centre Number

Candidate Number

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**

General Certificate of Education Advanced Level

CHEMISTRY**9189/2**

PAPER 2

JUNE 2017 SESSION**1 hour 15 minutes**

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

This question paper consists of 8 printed pages.

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[Turn over

- 1 (a) (i) Describe the bonding and structure in graphite.

- (ii) State and explain **one** property of graphite in terms of its structure and bonding.

property _____

explanation: _____

[5]

- (b) (i) State **two** assumptions of the kinetic theory in relation to an ideal gas.

1. _____
2. _____

- (ii) State the ideal gas equation.

- (iii) A gas weighs 2.00 g and occupies a volume of 1.714 dm^3 at 103 Pa and 23°C .

Calculate the molar mass of the gas.

molar mass = _____

- (iv) Assuming that one of the elements in the gas in (iii) is carbon, suggest the identity of the gas.

[6]

[Total: 11]

2

- (a) (i) Define the term *standard enthalpy of neutralisation*.

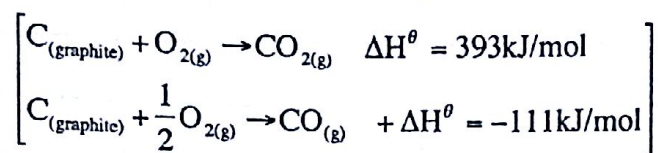
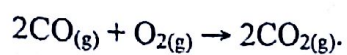
- (ii) When 500 cm^3 of 2M NaOH were added to 500 cm^3 of 2M HCl in a polythene beaker, the temperature rise was 13°C .

Calculate the enthalpy of neutralisation.

[Specific heat capacity of water is $4,2 \text{ J/kg}^\circ\text{C}$]

enthalpy of neutralization = _____ [4]

- (b) (i) Construct an energy cycle to determine the standard enthalpy change, ΔH^θ , for the reaction



- (ii) Use the cycle in (i) to determine the enthalpy change for the reaction.

$\Delta H^\theta =$ _____

[4]

[Total: 8]

- 3 (a) State and explain the trend in volatility of chlorine, bromine and iodine.

[2]

- (b) (i) Write a chemical equation to show the reaction of chlorine with

1. cold potassium hydroxide,

2. hot potassium hydroxide.

- (ii) Deduce the change in oxidation state in chlorine when it reacts with

1. cold potassium hydroxide,

2. hot potassium hydroxide.

[4]

- (c) (i) Describe and explain the observations made when chlorine is bubbled into a solution of silver bromide.

- (ii) Write the chemical equation for the reaction in (i).

[3]

- (d) An antiseptic cream is suspected to contain iodide ions as an active ingredient.

Describe a reaction that can be used to show the presence of iodide ions in the cream.

[2]

[Total: 11]

- 4 (a) Fig. 4.1 shows the structure of an organic compound B.

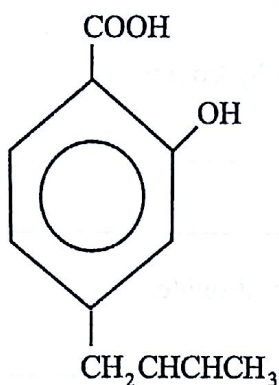


Fig. 4.1

- (i) State any **two** functional groups in compound B.

1. _____

2. _____

- (ii) State the type of isomerism shown by compound B.

- (iii) Draw the isomers of compound B.

[5]

- (b) Draw the structure(s) of the organic product(s) formed when compound B reacts with

(i) PCl_5 ,

(ii) hot acidified KMnO_4 ,

(iii) dilute nitric acid.

[4]

[Total: 9]

5

(a) Fig. 5.1 shows how Phenylamine can be prepared from benzene.

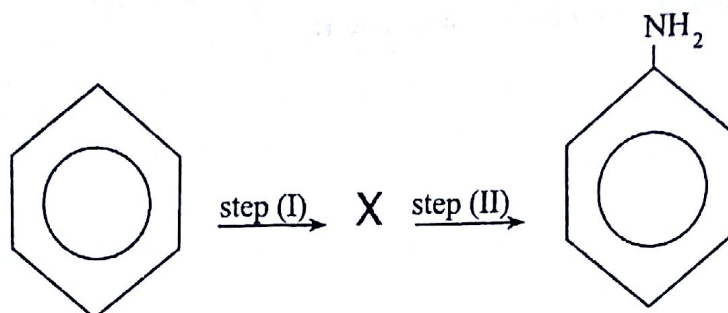


Fig. 5.1

State the reagents and conditions for

(i) Step 1, _____

(ii) Step II, _____

(iii) Deduce the structure of compound X.

[5]

(b) Compare the basicity of phenylamine to that of ethylamine.

[2]

(c) State the observation made when phenylamine reacts with bromine.

[1]

(d) Name the reagent used to convert ethylamine into an amide.

[1]

[Total: 9]